

ALI AL SALEM AB, Kuwait

WMO: 405736

Lat: 29.347N

Lon: 47.521E

Elev: 144

StdP: 99.61

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	5.8	-11.9	1.4	18.9	-10.0	1.6	21.8	11.5	16.5	10.3	16.5	3.7	280	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	48.9	21.4	47.8	20.9	46.3	20.4	27.0	37.0	25.0	37.1	23.0	37.4	6.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.3	34.4	21.4	16.3	32.2	18.8	13.9	29.2	86.0	38.0	76.9	38.4	68.8	36.1	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.6	9.4	DB	1.5	50.1	1.3	1.2	0.6	51.0	-0.2	51.6	-0.9	52.3	-1.8	53.1	(4)
(5)			WB	-1.4	27.8	2.2	1.7	-3.0	29.0	-4.3	30.0	-5.5	30.9	-7.1	32.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	13.3	15.6	20.2	26.8	33.1	37.0	38.6	38.5	34.7	29.4	20.0	14.9		
	DBStd	9.73	3.00	3.36	3.76	3.52	3.20	2.16	1.95	1.70	2.37	3.13	3.96	3.23		
	HDD10.0	11	6	2	0	0	0	0	0	0	0	0	0	3		
	HDD18.3	403	156	87	22	1	0	0	0	0	0	0	26	111		
	CDD10.0	6179	109	158	317	503	716	811	886	883	742	600	299	156		
	CDD18.3	3531	2	10	80	253	457	561	628	625	492	342	76	6		
	CDH23.3	58542	11	77	798	3105	7431	10264	11710	11504	8388	4563	645	45		
(13)	CDH26.7	42283	1	17	300	1668	5137	7874	9232	9026	6066	2733	227	3		
(14)	Wind	WSAvg	4.7	4.1	4.5	4.6	4.5	4.7	6.0	6.0	5.1	4.7	4.1	4.0	4.0	(14)
(15) Precipitation	PrecAvg	111	23	11	16	10	3	0	0	0	0	3	18	22		
	PrecMax	209	95	69	64	45	25	1	1	1	11	38	115	68		
	PrecMin	36	1	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	45	23	15	17	11	6	0	0	0	2	7	27	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	29.0	37.2	41.8	47.9	49.9	50.9	50.1	47.1	42.1	34.8	26.5		
		MCWB	13.5	13.9	15.4	21.0	24.5	24.2	21.9	21.7	21.0	19.4	17.0	13.5		
	2%	DB	22.1	25.2	32.9	38.2	45.0	47.8	49.0	48.2	45.2	40.1	31.2	24.8		
		MCWB	12.4	13.6	15.0	19.2	21.2	20.8	20.9	21.1	19.8	18.4	16.9	13.3		
	5%	DB	20.8	23.8	29.8	36.2	43.0	46.2	47.8	47.2	43.9	38.9	29.1	23.0		
		MCWB	12.3	12.8	14.6	17.8	19.6	20.2	20.4	20.8	19.4	18.1	15.5	13.3		
	10%	DB	19.0	22.0	27.2	34.2	41.0	44.9	46.1	46.1	42.2	37.1	27.1	21.1		
		MCWB	11.9	12.2	13.9	17.1	19.0	19.7	20.2	20.4	18.9	17.7	15.1	12.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	17.6	18.9	23.5	28.2	28.9	27.3	28.9	27.1	25.5	22.6	18.2		
		MCDB	19.4	20.4	24.6	37.1	45.8	47.3	37.0	37.8	33.0	30.0	26.0	20.7		
	2%	WB	15.8	16.4	17.8	21.0	24.2	24.5	24.7	27.2	22.9	23.8	20.6	16.8		
		MCDB	18.4	20.7	24.5	32.3	38.9	43.2	39.7	35.4	34.9	28.3	25.5	20.1		
	5%	WB	14.5	15.3	16.7	19.7	21.6	21.3	22.3	25.1	20.8	22.1	18.7	15.5		
		MCDB	17.7	20.1	24.6	30.5	35.6	42.6	43.9	38.4	41.2	28.8	23.8	19.1		
	10%	WB	13.0	14.1	15.4	18.8	20.2	19.8	20.9	22.4	19.8	20.4	17.4	14.1		
		MCDB	17.2	18.8	25.4	29.9	35.7	44.0	45.1	42.8	40.3	30.1	23.2	19.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.0	12.9	13.4	14.3	15.3	15.3	15.5	15.0	13.8	11.4	11.1		
		MCDBR	11.9	12.9	15.9	15.9	17.7	17.0	17.0	16.6	16.0	15.9	13.3	12.7		
		MCWBR	5.7	5.4	6.2	6.1	7.0	6.4	5.9	5.8	5.7	6.3	6.0	6.6		
	5% WB	MCDBR	9.3	10.1	11.8	13.7	16.2	17.5	17.0	15.5	15.6	12.4	10.3	9.3		
		MCWBR	5.7	5.6	5.8	5.9	8.3	8.5	7.6	8.2	6.5	7.2	6.0	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.495	0.543	0.600	0.690	0.695	0.648	0.662	0.613	0.563	0.567	0.537	0.498		
	taud		1.786	1.689	1.573	1.467	1.466	1.514	1.514	1.600	1.686	1.683	1.732	1.792		
	Ebn at Noon		703	710	703	661	662	690	679	707	724	679	656	674		
	Edh at Noon		185	222	266	305	306	290	290	263	234	220	194	177		
(44) All-Sky Solar Radiation	RadAvg		3.55	4.43	5.57	6.09	7.00	7.84	7.54	7.17	6.36	5.02	3.69	3.27		
	RadStd		0.25	0.26	0.22	0.30	0.31	0.26	0.20	0.15	0.20	0.21	0.29	0.29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.39	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	+129	+136

Nomenclature: See separate page